

Chandrakanth Peravelli

Staff Software Engineer

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Professional Summary

Staff Software Engineer with 14+ years of experience architecting and scaling distributed systems, microservices infrastructure, and data-intensive platforms. Backend and distributed-systems specialist focused on high availability, fault tolerance, and cloud architecture. Proven track record designing end-to-end solutions for complex operational challenges; skilled in event-driven and microservices patterns, cost optimization, and leading cross-functional teams through ambiguous, high-stakes initiatives. Hands-on generative AI engineer: built agentic systems with LangGraph, a Java MCP tool server (Spring AI), tool calling, RAG on pgvector, guardrails, human-in-the-loop approval, and automated agent evals.

Work Experience

- Staff Software Engineer – Cloudera, Chaska, Minnesota (Remote)** **Aug 2024 – Present**
 - Own architecture and technical strategy for Apache Atlas, a mission-critical enterprise metadata-governance platform serving large-scale global customer deployments; established reliability and scalability frameworks defining standards across multiple teams.
 - Architected AMRA (Atlas Modular Runtime Architecture), a next-generation distributed runtime framework that increased platform data throughput by 3x, made service startup 2x faster, and eliminated legacy infrastructure constraints for enterprise-scale metadata workloads.
 - Led platform reliability transformation that re-architected failover and continuous-availability logic, reducing unplanned downtime by over 95% and establishing the canonical pattern for all future resiliency initiatives.
 - Re-architected cloud-to-cloud metadata replication from a blocking, all-or-nothing synchronous job into an asynchronous pipeline with checkpointing, resume, automatic retry, concurrent multi-job submission, and snapshot-level observability; made replication 5x faster for jobs of 10,000+ entities with lineage while preserving strict data integrity across regions.
 - Translated organizational reliability objectives into a multi-quarter delivery roadmap across multiple engineering teams; kept execution on schedule through mid-cycle scope changes.
 - Mentor engineers on distributed-systems design patterns, establish architectural standards, and drive performance optimizations across previously siloed teams.
- Lead Engineer – Target Corporation, Minneapolis, MN** **April 2018 – Aug 2024**
 - Led architecture and delivery of enterprise 3D automation pipeline processing 1M+ daily transactions, integrating Airflow, Node.js, Python, Kafka, and event-driven microservices; generated \$2M in combined cost savings for Target and vendor partners.
 - Architected fault-tolerant microservices ecosystem spanning Kafka, AWS S3, GCP, Spring Boot, Kotlin, Postgres, and MongoDB; scaled system to handle 1,000+ requests/second during peak demand periods.
 - Engineered algorithmic solutions for guest order fulfillment optimization using PySpark, Python, Spring Boot, and Kafka, reducing fulfillment time and delivering \$1M+ in annual operational savings.
 - Developed enterprise security framework standardizing OAuth token validation and credential management across Java, Python, and Node.js ecosystems; secured 50+ internal and partner APIs.
 - Migrated product search from Postgres-based search to Elasticsearch, adding fuzzy matching, exact search, and autocomplete; improved search relevance and response times for guest product discovery.
 - Implemented automated functional and performance testing in CI/CD pipelines, reducing deployment cycle time and enabling teams to maintain code quality at velocity.
 - Mentored 8+ engineers on architectural patterns and design trade-offs; established code review standards and fostered culture of technical rigor.
- Senior Software Engineer – Best Buy, Minneapolis, Minnesota** **May 2015 – April 2018**
 - Built Best Buy Enterprise Microservices Library, a reusable framework eliminating API redundancy across 30+ applications; reduced developer effort by 30% and accelerated microservice delivery.
 - Established REST API governance using OpenAPI standards; created consistency framework enabling seamless integration across internal and partner ecosystems.

- Delivered custom observability dashboards in Grafana and Kibana for real-time system health and error analysis; enabled proactive incident detection and root-cause investigation.
- Optimized database performance using PostgreSQL and jOOQ query patterns; reduced query latency and improved system responsiveness under high concurrent load.

● Senior Associate | Projects – Tata Consultancy Services, Chennai, India

Nov 2011 – May 2015

- Designed and coded enterprise application solutions using Spring MVC, Hibernate, and jQuery; delivered software meeting functional requirements across global client base.
- Implemented continuous data pipelines transferring archive data to AWS S3; enabled downstream analytics and reporting workflows while ensuring data durability and compliance.

Generative Ai Projects

github.com/jackhalfalltrades/Gen-AI

Sentinel – Agentic Incident & Security Investigation for Retail Operations | Python, Java

- Built a LangGraph agent that takes an alert, routes it to an incident or security workflow, attaches a playbook, investigates through tools, and writes a root cause that waits for human approval (LangGraph interrupt with Postgres checkpointing, so cases survive restarts).
- Built a Java MCP server (Spring AI 2.0, Spring Boot) exposing four tools (logs, metrics, data lineage, IAM) over a Postgres event store, keeping the agent decoupled from the data source.
- Modeled a simulated retailer with six seeded faults (DB pool exhaustion, Kafka consumer lag, bad promo config, credential stuffing, expired service account, PCI table read); eval harness identifies 6/6 root causes.
- FastAPI service to investigate, approve, and list cases; multi-provider LLM layer (OpenAI, Anthropic, Gemini, Ollama) switched through configuration.

Digital Twin – Interactive Resume Agent | Python

- Built a recruiter-facing agent that answers only from my resume and LinkedIn profile using RAG on Postgres + pgvector (chunking, OpenAI embeddings, top-k retrieval).
- Async LangGraph state machine with tool calling, an LLM input guardrail, and an output groundedness judge with safe fallback; structured-output nodes for skills and experience.
- Long-conversation memory using a sliding window, LLM compaction, and vector recall of older turns; streaming Gradio UI, Dockerized.

Education

Bachelor of Engineering in Information Technology – Anna University, India

May 2011

Awards & Achievements

- Received 2 engineering recognition awards at Cloudera (2024–present).
- Received 3 engineering recognition awards at Target Corporation (2018–2024).
- Delivered a 3x throughput increase (AMRA) and 95% reduction in unplanned downtime on Apache Atlas at Cloudera.
- Generated \$3M+ in combined cost savings at Target through 3D automation and order-fulfillment optimization.

Skills

- **Core Languages & Frameworks:** Java, J2EE, Spring Boot, Kotlin, Kotlin Coroutines, Python, FastAPI, Node.js, TypeScript
- **Generative AI & Agentic Systems:** LangGraph, LangChain, OpenAI Agents SDK, Model Context Protocol (MCP), Spring AI, Tool Calling, RAG, pgvector, Embeddings, Context Engineering, Guardrails / LLM-as-Judge, Human-in-the-Loop, Agent Evaluation, OpenAI, Anthropic, Gemini, Ollama, Hugging Face, Gradio
- **Architecture & Design:** Distributed Systems, High Availability, Microservices, Event-Driven Architectures, REST APIs, CQRS, Design Patterns, OAuth, JWT
- **Data & Streaming:** Kafka, Apache Atlas, Metadata Management, Elasticsearch, Redis, MongoDB, Postgres, DynamoDB, SQL Server, Oracle, jOOQ, ZooKeeper, RocksDB, Cassandra
- **Cloud & Infrastructure:** AWS (S3, Kinesis, SNS, SQS, API Gateway, DynamoDB), Google Cloud Platform (GCP), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins, Spinnaker, Vela)
- **Observability & Quality:** Grafana, Kibana, Splunk, ELK Stack, InfluxDB, MockK, JUnit, TestNG, Spock, K6, Cucumber, performance testing
- **Process & Collaboration:** Agile, JIRA, Git, technical mentoring, code review